

LILAVATI MHASKE

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SUMMARY

Software Engineer and MSc Advanced Computer Science student at the University of Leicester with strong foundations in Object-Oriented Programming, Computer Networks, and Database Management Systems. Experienced in building and testing backend and distributed systems using Java and Python, with hands-on exposure to system architecture, debugging, and reliability engineering. Seeking software engineering roles where strong computer science fundamentals and scalable system design are valued.

TECHNICAL SKILLS

Major Competencies: Software Development, Backend Engineering, Distributed Systems, System Design (Basic), Debugging & Reliability Engineering

Programming Languages: Java, Python, JavaScript, C, C++

Backend & Web Technologies: Node.js, Express.js, RESTful APIs, HTTP/HTTPS

Computer Science Fundamentals: Object-Oriented Programming (OOP), Data Structures & Algorithms, Computer Networks, Database Management Systems (DBMS), Operating Systems

Databases: MySQL, MongoDB, Oracle, SQL

Cloud & Infrastructure: AWS (EC2, VPC, S3, Lambda), NGINX, Redis (Pub/Sub)

Tools & Platforms: Git, GitHub, Linux, CI/CD Pipelines, Prometheus, Grafana, Wireshark, VS Code, IntelliJ IDEA

Software Development Methodologies: Agile Development, Test-Driven Development (TDD)

AI & Data: Machine Learning, Generative AI (Foundational)

EDUCATION

2025 - 2026 M.S. Advanced Computer Science, University of Leicester

2021 - 2024 B. E. Computer Science and Engineering, Savitribai Phule Pune University, India

2018 - 2021 Diploma in Computer Science Engineering, Savitribai Phule Pune University, India

Work Experience

Amazon

Jan 2024 – Jun 2024

Quality Assurance Engineer Intern

Chennai, India

- Developed and optimised Python-based automation frameworks to test autonomous robotics software, improving test execution efficiency by ~20%.
- Designed and executed system-level test cases for autonomous navigation, precision movement, obstacle avoidance, and docking behaviours.
- Conducted regression and exploratory testing to identify edge cases impacting robot navigation, recovery logic, and system reliability in dynamic environments.
- Analysed inconsistent behaviour across simulation and CI pipeline environments using logs, metrics, and behavioural traces to improve system stability.
- Designed test scenarios and data collection workflows to validate autonomous recovery mechanisms when navigation paths were obstructed.
- Collaborated closely with software engineers to debug failures, validate fixes, and improve release reliability by reducing navigation failure rates.

NoQs

July 2023 – Nov 2023

Web Developer Intern

Remote

- Designed and developed responsive, user-facing web applications using HTML, CSS, JavaScript, and React, improving accessibility, usability, and overall user experience.
- Built and integrated RESTful backend APIs using Node.js and Express, reducing backend response time by ~15% through efficient routing and request handling.

- Performed end-to-end testing and systematic debugging across frontend and backend components, reducing application errors by ~20%.
 - Collaborated within an Agile development environment, contributing to faster feature delivery, iterative improvements, and improved code quality.
 - Optimised database queries and implemented secure data-handling practices using MongoDB, improving performance for high-traffic scenarios.
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PROJECTS

Distributed Whiteboard Application | Node.js, WebSockets, Redis, AWS, NGINX, Prometheus, Grafana

- Designed and developed a distributed, real-time collaborative system from scratch, enabling multiple users to interact concurrently with low latency.
- Architected backend services deployed on AWS EC2 within a VPC, using NGINX as a reverse proxy for traffic management.
- Implemented Redis Pub/Sub to support scalable, event-driven message propagation across distributed nodes with fault tolerance.
- Integrated monitoring and observability using Prometheus, Grafana, and CloudWatch to track latency, throughput, and system health.
- Conducted scalability, concurrency, and failure-mode testing, identifying bottlenecks and improving system reliability.

Integrity-Driven Voting Platform | React, FastAPI, MySQL, REST APIs

- Designed and developed a full-stack web platform enabling structured, data-driven decision-making through digital referendums and surveys.
 - Built secure RESTful APIs to manage users, votes, and results, ensuring data integrity, consistency, and correctness.
 - Designed and normalized relational database schemas in MySQL to support user participation, vote aggregation, and result analytics.
 - Implemented backend validation, exception handling, logging, and error reporting to improve system reliability.
 - Focused on scalable architecture, clean API design, and maintainable backend services aligned with real-world governance use cases.
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LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Student Union Member University of Leicester (Leicester 100 Programme)

- Selected among high-performing students to represent the university in leadership, professional development, and community engagement initiatives.

Training & Placement Team Head Information Technology Student's Association

- Collaborated with the Training and Placement Department to support student career development initiatives.

Event Management Team Lead Commit Association, Government Polytechnic Pune

- Led a team of 5 members to plan and execute campus events, ensuring effective coordination and timely delivery.
- Delegated responsibilities and managed logistics to ensure smooth execution of events.

President Tree Plantation Drive

- Organised and participated in tree plantation activities to promote environmental sustainability.
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ACHIEVEMENTS & AWARDS

- **First Prize – Software Development Methods Competition (ESTECO):** Recognised for strong problem-solving skills and application of software engineering principles.
- **Student of the Year Award:** Awarded for academic excellence, leadership, and active involvement in extracurricular activities.
- **1st Place – ASPIRE-2K24 National Project Competition:** Developed an innovative drone-based pollution detection system using computer vision.
- **2nd Prize – NES Innovation Awards 2024:** Recognised for applied AI research in environmental monitoring.